

1 Fundamentals: Logic, sets, numbers and vectors

1.1 Real Numbers, Eulidean Spaces

Axioms of Addition:

- Axioms:

- A1 (Associativity)** $x + (y + z) = (x + y) + z \quad \forall x, y, z \in \mathbb{R}$
- A2 (Neutral element)** $x + 0 = x \quad \forall x \in \mathbb{R}$
- A3 (Inverse element)** $x + (-x) = 0 \quad \forall x \in \mathbb{R} \exists (-x) \in \mathbb{R}$
- A4 (Commutativity)** $x + y = y + x \quad \forall x, y \in \mathbb{R}$

Axioms of Multiplication:

- Axioms:

- M1 (Associativity)** $x \cdot (y \cdot z) = (x \cdot y) \cdot z \quad \forall x, y, z \in \mathbb{R}$
- M2 (Neutral element)** $x \cdot 1 = x \quad \forall x \in \mathbb{R}$
- M3 (Inverse element)** $x \cdot x^{-1} = 1 \quad \forall x \in \mathbb{R} \setminus \{0\} \exists x^{-1} \in \mathbb{R}$
- M4 (Commutativity)** $x \cdot z = z \cdot x \quad \forall x, z \in \mathbb{R}$

Distributivity (Compatibility of addition and mult.):

- Axiom:

$$\text{D} \quad x \cdot (y + z) = xy + xz = (y + z) \cdot x \quad \forall x, y, z \in \mathbb{R}$$

Order Axioms:

- Axioms:

- O1 (Reflexivity)** $x \leq x \quad \forall x \in \mathbb{R}$
- O2 (Transitivity)** $(x \leq y \wedge y \leq z) \Rightarrow x \leq z$
- O3 (Antisymmetry)** $(x \leq y \wedge y \leq x) \Rightarrow x = y$
- O4 (Totality)** $\forall x, y \in \mathbb{R} : x \leq y \text{ or } y \leq x$

Compatibility (Consistency with addition and mult.):

- Axioms:

- K1** $x \leq y \Rightarrow x + z \leq y + z \quad \forall x, y, z \in \mathbb{R}$
- K2** $x \geq 0, y \geq 0 \Rightarrow xy \geq 0 \quad \forall x, y \in \mathbb{R}$

R 1.1.5:

- The set $\mathbb{Q}$ of rational numbers, equipped with addition, multiplication, and the order relation $\leq$, satisfies the above axioms.

V (Completeness axiom):

- The set $\mathbb{R}$ is **order complete**: For any two nonempty sets $A, B \subset \mathbb{R}$ such that

$$a \leq b \quad \text{for all } a \in A, b \in B,$$

there exists a number $c \in \mathbb{R}$ satisfying

$$a \leq c \leq b \quad \forall a \in A, b \in B.$$

C 1.1.6:

- The consequences of above axioms are:

1. Uniqueness of the additive and multiplicative inverse.
2. $0 \cdot x = 0 \quad \forall x \in \mathbb{R}$.
3. $(-1) \cdot x = -x \quad \forall x \in \mathbb{R}$, in particular $(-1)^2 = 1$.
4. $y \geq 0 \iff (-y) \leq 0$.
5. $y^2 \geq 0 \quad \forall y \in \mathbb{R}$, in particular $1 = 1 \cdot 1 \geq 0$.
6. If $x \leq y$ and $u \leq v$, then $x + u \leq y + v$.
7. If $0 \leq x \leq y$ and $0 \leq u$, then $xu \leq yu$.

T (Completeness of the real numbers):

- $\mathbb{R}$ is a commutative ordered field that is order-complete.

C 1.1.7 (Archimedean Principle):

- 1. For $x \in \mathbb{R}$ and $y > 0$ there exists $n \in \mathbb{N}$ such that

$$ny > x.$$

2. For every $\varepsilon > 0$ there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < \varepsilon.$$

T 1.1.8:

- For every $t \geq 0$, $t \in \mathbb{R}$, the equation $x^2 = t$ has a solution in $\mathbb{R}$.

D (Bounded set - upper/lower bounded):

- A set $A \subset \mathbb{R}$ is called upper bounded if

$$\exists b \in \mathbb{R} \quad \forall a \in A : a \leq b.$$

Each such b is called an upper bound for A . (Similarly for lower bound.)

T (Supremum Infimum):

- 1. Maximum and minimum (if exists) are unique
- 2. Every non-empty subset bounded below (above) has a unique infimum (supremum).
- 3. $S = \sup(X) \iff$

$$(\forall x \in X : x \leq S) \wedge (\forall \varepsilon > 0 \exists x \in X : x > S - \varepsilon)$$

4. $I = \inf(X) \iff$

$$(\forall x \in X : x \geq I) \wedge (\forall \varepsilon > 0 \exists x \in X : x < I + \varepsilon)$$

D (Maximum):

- Maximum is:

$$\max\{x, y\} := \begin{cases} x, & \text{if } y \leq x, \\ y, & \text{if } x \leq y. \end{cases}$$

D (Minimum):

- Minimum is:

$$\min\{x, y\} := \begin{cases} y, & \text{if } y \leq x, \\ x, & \text{if } x \leq y. \end{cases}$$

T (Properties of the absolute value):

- Properties:

1. $|x| \geq 0$.
2. $|x + y| \leq |x| + |y|$ (triangle inequality).
3. $|x + y| \geq ||x| - |y||$ (inv. triangle inequality).

T (Young's inequality):

- For every $\varepsilon > 0$ and all $x, y \in \mathbb{R}$ it holds:

$$2|xy| \leq \varepsilon x^2 + \frac{1}{\varepsilon} y^2.$$

1.2 Vectors and Vector Spaces

D (inner product):

- For $x, y \in \mathbb{R}^n$ the standard inner product is

$$x \cdot y = \langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

D (Euclidean norm):

- The Euclidean norm is

$$\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$$

D (Euclidean distance):

- The Euclidean distance is

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

D (Triangle inequality):

- For all $x, y, z \in \mathbb{R}^n$:

$$\|x - z\| \leq \|x - y\| + \|y - z\|$$

1.3 Complex Numbers

D (Complex number):

- A complex number is

$$z = a + ib, \quad a, b \in \mathbb{R}$$

D (set of complex numbers):

- The set of complex numbers is

$$\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$$

D (Terminology of complex numbers):

- Terminology
 - $a = \operatorname{Re}(z)$ — real part
 - $b = \operatorname{Im}(z)$ — imaginary part
 - $i^2 = -1$
 - If $a = 0$, the number is purely imaginary

D (Complex numbers Addition and Subtraction):

- Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$.

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$$

D (Complex numbers Multiplication):

- Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$.

$$z_1 z_2 = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + y_1 x_2)$$

D (Complex conjugate):

- For a complex number $z = x + iy$, the complex conjugate of z is given by

$$\bar{z} = \overline{x + iy} = x - iy.$$

D (Complex numbers Division):

- To divide numbers we use conjugate:

$$\frac{z}{w} = \frac{z\bar{w}}{w\bar{w}} = \frac{z\bar{w}}{|w|^2}$$

D (Properties of the Conjugate):

- Properties are:

- $z\bar{z} = |z|^2$
- $z + \bar{z} = 2\operatorname{Re}(z)$, $z - \bar{z} = 2i\operatorname{Im}(z)$
- $\bar{\bar{z}} = z$
- $\overline{z \pm w} = \bar{z} \pm \bar{w}$
- $\overline{z\bar{w}} = \bar{z}w$
- $z = \bar{z}$, if $z \in \mathbb{R}$
- $\bar{z} = -z$, z is purely imaginary

D (Modulus):

- The magnitude $|z|$ of a complex number $z = a + ib$ is the length of the corresponding vector.

$$|z| = \sqrt{a^2 + b^2}$$

The distance between two complex numbers z_1 and z_2 is

$$d(z_1, z_2) = |z_2 - z_1|$$

Triangle inequality works also for complex numbers:

$$|z + w| \leq |z| + |w|$$

D (Euler's Formula):

- The exponential function can be represented by a series expansion as

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

If we substitute $x = it$ into this formula, we obtain Euler's formula.

$$e^{it} = \cos t + i \sin t$$

We can write $\sin(t)$ and $\cos(t)$ using complex exponential functions.

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}$$

$$\sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

D (Exponential with Complex Argument):

- We can also allow complex arguments for the exponential function. Then the following rules apply:

- $e^{z+w} = e^z e^w$
- $e^{a+ib} = e^a (\cos b + i \sin b)$
- $e^{z+i2\pi} = e^z$

D (Polar Form):

- A complex number can also be described by specifying the distance from the origin and by specifying a suitable angle (**polar form**):
 $z = re^{i\varphi}$.

- $r = |z| \geq 0$ - distance
- $\varphi \in (-\pi, \pi]$ - polar angle
- $\varphi = \arg(z) = \arccos(z)$

D (Polar to Cartesian conversion):

- Polar to Cartesian conversion is:

- $x = r \cos \varphi$
- $y = r \sin \varphi$

D (Cartesian to Polar conversion):

- Cartesian to Polar conversion is:

- $r = \sqrt{x^2 + y^2}$
- $\varphi = \arctan \frac{y}{x}$

If $x = 0$:

$$\arg(iy) = \begin{cases} \frac{\pi}{2}, & y > 0 \\ -\frac{\pi}{2}, & y < 0 \end{cases}$$

D (Powers of Complex Numbers):

- The following rules must be observed:

- $z_1 z_2 = r_1 r_2 e^{i(\varphi_1 + \varphi_2)}$
- $\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\varphi_1 - \varphi_2)}$
- $z^n = r^n e^{in\varphi}$

D (Roots from complex numbers):

- For $n \in \mathbb{N}$, the solutions to the equation $z^n = re^{i\varphi}$ are

$$z_k = r^{1/n} e^{i(\varphi/n + 2\pi k/n)}, \quad k = 0, \dots, n-1$$

D (Quadratic Equations):

- For the quadratic equation

$$az^2 + bz + c = 0,$$

in complex numbers the solution is still

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

D (Fundamental Theorem of Algebra):

- Each polynomial

$$p(z) = \sum_{k=0}^n a_k z^k, \quad a_k \in \mathbb{C}$$

can be factored into linear factors, i.e. written as

$$p(z) = a_n (z - z_1)(z - z_2) \dots (z - z_n)$$

The numbers z_k are therefore precisely the zero points of $p(z)$ (with **multiplicity**).

C (non-real roots appear in conjugate pairs):

- Non-real roots of a polynomial with **real** coefficients occur in conjugate pairs $z, \bar{z}$.

2 Sequences

D (Sequence):

• A sequence $(a_n)_{n \in \mathbb{N}_0}$ (also written $(a_n)_{n \geq 0}$ or $(a_n)_{n=0}^\infty$) is an infinite list of numbers, where each entry is called a *term* of the sequence. A sequence can also be viewed as a function

$$f : \mathbb{N}_0 \rightarrow \mathbb{R} \quad (\text{or } f : \mathbb{N} \rightarrow \mathbb{R}),$$

whose values are $f(n) = a_n$.

D (Ways to Describe a Sequence):

• One way to describe a sequence is by giving an explicit formula for each term:

$$a_n = f(n).$$

Another possibility is to define the sequence recursively, constructing each new term from previous ones:

$$a_n = g(a_{n-1}, a_{n-2}, \dots).$$

The functions f and g are called the *rule of formation* of the sequence.

D (Convergence/Divergence):

• A sequence $(a_n)_{n \in \mathbb{N}_0} \subset \mathbb{R}$ is called *convergent* if there exists a number $L \in \mathbb{R}$ such that

$$\forall \varepsilon > 0 \exists N > 0 \text{ such that } \forall n > N : |a_n - L| < \varepsilon.$$

The number L is called the *limit* of the sequence, and we write

$$\lim_{n \rightarrow \infty} a_n = L.$$

If no such L exists, the sequence is called *divergent*.

T (Uniqueness of the Limit):

• A convergent sequence has exactly one unique limit.

D (Divergence to infinity):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence.

• If

$$\forall M > 0 \exists N > 0 \text{ such that } \forall n > N : a_n > M,$$

then the sequence diverges to infinity and we write

$$\lim_{n \rightarrow \infty} a_n = \infty.$$

• If

$$\forall M > 0 \exists N > 0 \text{ such that } \forall n > N : a_n < -M,$$

then the sequence diverges to minus infinity and we write

$$\lim_{n \rightarrow \infty} a_n = -\infty.$$

D (Subsequence):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A *subsequence* is a sequence of the form

$$(a_{n_k})_{k \in \mathbb{N}_0},$$

where $(n_k)_{k \in \mathbb{N}_0}$ is a sequence of non-negative integers satisfying

$$n_{k+1} > n_k \quad \text{for all } k \in \mathbb{N}_0.$$

L (Limit of a sequence and its subsequence):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a convergent sequence with limit $L \in \mathbb{R}$. Then every subsequence $(a_{n_k})_{k \in \mathbb{N}_0}$ also converges to the same limit L .

D (Accumulation Point):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is called an accumulation point of the sequence if

$$\forall \varepsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N \text{ such that } |a_n - A| < \varepsilon.$$

Intuitively, every interval $(A - \varepsilon, A + \varepsilon)$ contains terms a_n for infinitely many indices n .

T (Characterization of Accumulation Points):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is an accumulation point of the sequence if and only if there exists a convergent subsequence $(a_{n_k})_{k \in \mathbb{N}_0}$ such that

$$\lim_{k \rightarrow \infty} a_{n_k} = A.$$

C (Accumulation Point of a Convergent Sequence):

• Every convergent sequence $(a_n)_{n \in \mathbb{N}_0}$ in $\mathbb{R}$ has exactly one accumulation point, which coincides with its limit.

C (Infinite Occurrence Near an Accumulation Point):

• Let A be an accumulation point of the sequence $(a_n)_{n \in \mathbb{N}_0}$. Then for every $\varepsilon > 0$ there exist infinitely many terms of the sequence lying in the interval $(A - \varepsilon, A + \varepsilon)$.

T (Algebra of Limits):

• Assume

$$\lim_{n \rightarrow \infty} a_n = K, \quad \lim_{n \rightarrow \infty} b_n = L$$

with $K, L \neq \pm\infty$ and let C be a constant. Then

$$\lim_{n \rightarrow \infty} (a_n + b_n) = K + L$$

$$\lim_{n \rightarrow \infty} (a_n - b_n) = K - L$$

$$\lim_{n \rightarrow \infty} (a_n b_n) = KL$$

$$\lim_{n \rightarrow \infty} (C a_n) = CK$$

If $L \neq 0$ and $b_n \neq 0$ then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{K}{L}.$$

If $K < L$ then there exists N such that $a_n < b_n$ for all $n \geq N$. If there exists N such that $a_n \leq b_n$ for all $n \geq N$, then $K \leq L$.

T (Important Limits):

•

$$\lim_{n \rightarrow \infty} \frac{\log n}{n} = 0 = \lim_{n \rightarrow \infty} \frac{\ln n}{n}$$

$$\lim_{n \rightarrow \infty} n^{1/n} = 1$$

$$\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$$

$$\forall x \in \mathbb{R} : \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

$$\forall x \in \mathbb{R} : \lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$$

T (Sandwich Theorem):

• Let $(a_n)_{n \in \mathbb{N}_0}$ and $(c_n)_{n \in \mathbb{N}_0}$ be convergent sequences with

$$\lim_{n \rightarrow \infty} a_n = L, \quad \lim_{n \rightarrow \infty} c_n = L$$

and suppose there exists N_0 such that

$$a_n \leq b_n \leq c_n \quad \text{for all } n \geq N_0.$$

Then $(b_n)_{n \in \mathbb{N}_0}$ is also convergent and

$$\lim_{n \rightarrow \infty} b_n = L.$$

D (Bounded Sequence):

• A sequence $(a_n)_{n \in \mathbb{N}_0}$ is called bounded below or bounded above if the set

$$M = \{a_n \mid n \in \mathbb{N}_0\}$$

has a lower bound or an upper bound respectively. A sequence that is bounded both above and below is called a bounded sequence.

D (Monotone Sequence):

• A sequence $(a_n)_{n \in \mathbb{N}_0}$ is called monotone decreasing if

$$m > n \Rightarrow a_m \leq a_n$$

and monotone increasing if

$$m > n \Rightarrow a_m \geq a_n.$$

It is called strictly monotone decreasing if

$$m > n \Rightarrow a_m < a_n$$

and strictly monotone increasing if

$$m > n \Rightarrow a_m > a_n.$$

T (Monotone Convergence Theorem):

- Every bounded and monotone sequence converges.

L (Convergent sequence = bounded):

- Every convergent sequence is bounded.

T Monotone Convergence:

- Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence of real numbers.

(i) A monotone sequence converges if and only if it is bounded.

(ii) If (a_n) is monotonically increasing, then

$$\lim_{n \rightarrow \infty} a_n = \sup\{a_n \mid n \in \mathbb{N}_0\}.$$

(iii) If (a_n) is monotonically decreasing, then

$$\lim_{n \rightarrow \infty} a_n = \inf\{a_n \mid n \in \mathbb{N}_0\}.$$

D Limit Superior and Limit Inferior:

- Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence.

- *limit superior* of the sequence

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup\{a_k \mid k \geq n\}$$

- *limit inferior* of the sequence.

$$\liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf\{a_k \mid k \geq n\}$$

L Properties of limsup and liminf:

- Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence. Then:

(i) $\limsup_{n \rightarrow \infty} a_n = \inf_{n \in \mathbb{N}} (\sup\{a_k \mid k \geq n\})$

(ii) $\liminf_{n \rightarrow \infty} a_n = \sup_{n \in \mathbb{N}} (\inf\{a_k \mid k \geq n\})$

(iii) If (a_n) is convergent, then

$$\lim_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n.$$

(iv) A bounded sequence (a_n) converges if and only if

$$\lim_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n.$$

T Characterization via limsup:

- Let $(a_n)_{n \in \mathbb{N}_0}$ be a bounded sequence and define

$$A = \limsup_{n \rightarrow \infty} a_n.$$

Then:

(i) A is an accumulation point of the sequence.

(ii) For every $\varepsilon > 0$, there are only finitely many indices n such that

$$a_n > A + \varepsilon.$$

(iii) For every $\varepsilon > 0$, there are infinitely many indices n such that

$$A - \varepsilon < a_n < A + \varepsilon.$$

An analogous statement holds for the limit inferior.

C Subsequence of a Bounded Sequence:

- Every bounded sequence of real numbers has at least one accumulation point and admits a convergent subsequence.

D Cauchy Sequence:

- A sequence $(a_n)_{n \in \mathbb{N}_0} \subset \mathbb{R}$ is called a *Cauchy sequence* if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |a_n - a_m| < \varepsilon.$$

L Cauchy Sequences are Bounded:

- Every Cauchy sequence is bounded.

T Cauchy Criterion:

- A sequence of real numbers converges if and only if it is a Cauchy sequence.

3 Series

D Series:

- A (formal) expression of the form

$$a_0 + a_1 + a_2 + \cdots = \sum_{n=0}^{\infty} a_n$$

is called a *series*. The terms a_n are called elements or summands. If the sequence of partial sums

$$s_n := \sum_{k=0}^n a_k$$

converges to a limit $L \in \mathbb{R}$, then the series is said to *converge*, and we write

$$\sum_{n=0}^{\infty} a_n = L.$$

Otherwise, the series is called *divergent*.

T Necessary Condition for Convergence:

- If the series $\sum_{n=0}^{\infty} a_n$ converges, then necessarily

$$\lim_{n \rightarrow \infty} a_n = 0.$$

L Linearity of Series:

- Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be convergent series. Then:

$$\begin{aligned} \sum_{n=0}^{\infty} (a_n + b_n) &= \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n, \\ \sum_{n=0}^{\infty} C a_n &= C \sum_{n=0}^{\infty} a_n \quad (C \in \mathbb{R}). \end{aligned}$$

L Tail of a Series:

- Let $\sum_{n=0}^{\infty} a_n$ be a series. For any $N \in \mathbb{N}_0$:

$$\sum_{n=0}^{\infty} a_n \text{ converges} \iff \sum_{n=N}^{\infty} a_n \text{ converges,}$$

and in that case

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{N-1} a_n + \sum_{n=N}^{\infty} a_n.$$

T Series with Non-Negative Terms:

- Let $a_n \geq 0$ for all n . Then the sequence of partial sums

$$s_n = \sum_{k=0}^n a_k$$

is monotonically increasing. If (s_n) is bounded, then the series $\sum_{n=0}^{\infty} a_n$ converges; otherwise, it diverges.

T Comparison Test:

- Let $\sum a_n, \sum b_n, \sum c_n$ be series with non-negative terms and suppose that for some N :

$$c_n \leq b_n \leq a_n \quad \text{for all } n \geq N.$$

Then:

- If $\sum a_n$ converges, then $\sum b_n$ converges (majorant test).
- If $\sum c_n$ diverges, then $\sum b_n$ diverges (minorant test).

D Absolute and Conditional Convergence:

- A series $\sum a_n$ is called *absolutely convergent* if $\sum |a_n|$ converges.
- It is called *conditionally convergent* if $\sum a_n$ converges but $\sum |a_n|$ does not.

T Riemann Rearrangement Theorem:

- Let $\sum a_n$ be a conditionally convergent series of real numbers, and let $L \in \mathbb{R}$. Then there exists a bijection $\varphi : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that

$$\sum_{n=0}^{\infty} a_{\varphi(n)} = L.$$

T Rearrangement of Absolutely Convergent Series:

- Let $\sum a_n$ be absolutely convergent and let $\varphi : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be a bijection. Then

$$\sum_{n=0}^{\infty} a_{\varphi(n)}$$

converges absolutely and

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} a_{\varphi(n)}.$$

T Leibniz Criterion:

- Let (a_n) be a monotonically decreasing sequence of non-negative real numbers with

$$\lim_{n \rightarrow \infty} a_n = 0.$$

Then the alternating series

$$\sum_{n=0}^{\infty} (-1)^n a_n$$

converges. Moreover, for all $m \in \mathbb{N}_0$:

$$\sum_{n=0}^{2m+1} (-1)^n a_n \leq \sum_{n=0}^{\infty} (-1)^n a_n \leq \sum_{n=0}^{2m} (-1)^n a_n.$$

T Cauchy Criterion for Series:

- The series $\sum_{n=0}^{\infty} a_n$ converges if and only if for every $\varepsilon > 0$ there exists N such that for all $n \geq m \geq N$:

$$\left| \sum_{k=m+1}^n a_k \right| \leq \varepsilon.$$

T Absolute Convergence Implies Convergence:

- Every absolutely convergent series converges, and

$$\left| \sum_{n=0}^{\infty} a_n \right| \leq \sum_{n=0}^{\infty} |a_n|.$$

T Root Test:

- Let (a_n) be a sequence and define

$$\rho = \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

Then:

- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

T Ratio Test:

- Let (a_n) with $a_n \neq 0$ and define

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

Then:

- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

T Cauchy Product:

- Let $\sum a_n$ and $\sum b_n$ be absolutely convergent. Then

$$\left(\sum_{n=0}^{\infty} a_n \right) \left(\sum_{n=0}^{\infty} b_n \right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_{n-k} b_k \right),$$

and the series on the right converges absolutely.

D Power Series:

- A series of the form

$$\sum_{k=0}^{\infty} c_k (x - a)^k$$

is called a *power series* centered at a with coefficients c_k .

D Convergence of a Power Series:

- For fixed x , if the sequence of partial sums

$$s_n(x) = \sum_{k=0}^n c_k (x - a)^k$$

converges, then the power series is said to converge at x .

T Radius of Convergence:

- Every power series has a radius of convergence R such that:

- The series converges for $|x - a| < R$,
- The series diverges for $|x - a| > R$.

The interval $(a - R, a + R)$ is called the interval of convergence. The radius R is given by:

$$\rho = \limsup_{n \rightarrow \infty} |c_n|^{1/n}.$$

Then

$$R = \begin{cases} 0 & \text{if } \rho = \infty, \\ \frac{1}{\rho} & \text{if } 0 < \rho < \infty, \\ \infty & \text{if } \rho = 0. \end{cases}$$

4 Functions

4.1 Function basics

D Function:

• A function (mapping) is a rule that assigns to each admissible input exactly one output in the target set. The set of all possible inputs is called the *domain*, denoted by $\text{domain}(f)$ or D . The set of all actually attained outputs is called the *image*, denoted by $\text{image}(f)$ or W .

$$f : \text{domain}(f) \rightarrow \text{range}(f), \quad x \mapsto f(x).$$

The input is called the *independent variable*, the output the *dependent variable*.

D Restriction of a Function:

• Let $f : D(f) \rightarrow \mathbb{R}$ and let $D_0 \subseteq D(f)$. The restriction of f to D_0 is the function

$$f|_{D_0} : D_0 \rightarrow \mathbb{R}, \quad f|_{D_0}(x) = f(x).$$

Note that f and $f|_{D_0}$ are, in general, different functions.

D Bounded Functions:

• Let $f : D(f) \rightarrow \mathbb{R}$.

• f is *bounded above* if there exists $M \in \mathbb{R}$ such that

$$f(x) \leq M \quad \forall x \in D(f).$$

• f is *bounded below* if there exists $M \in \mathbb{R}$ such that

$$f(x) \geq M \quad \forall x \in D(f).$$

• f is *bounded* if there exists $M \in \mathbb{R}$ such that

$$|f(x)| \leq M \quad \forall x \in D(f).$$

D Injective, Surjective, Bijective:

• Let $f : X \rightarrow Y$.

• f is *injective* if

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

• f is *surjective* if

$$\forall y \in Y \exists x \in X : f(x) = y.$$

• f is *bijective* if it is both injective and surjective.

D Composition:

• Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. The composition is defined by

$$g \circ f : X \rightarrow Z, \quad (g \circ f)(x) = g(f(x)).$$

D Identity and Inverse:

• For a set X , the identity function is defined by

$$\text{id}_X(x) = x \quad \forall x \in X.$$

If $f : X \rightarrow Y$ is bijective, then there exists a unique function

$$f^{-1} : Y \rightarrow X$$

such that

$$f^{-1} \circ f = \text{id}_X, \quad f \circ f^{-1} = \text{id}_Y.$$

This function is called the *inverse* of f .

D Monotonicity:

• Let $f : D(f) \subseteq \mathbb{R} \rightarrow \mathbb{R}$.

• f is *monotonically increasing* if

$$x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2).$$

• f is *strictly increasing* if

$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2).$$

• f is *monotonically decreasing* if

$$x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2).$$

• f is *strictly decreasing* if

$$x_1 < x_2 \Rightarrow f(x_1) > f(x_2).$$

T Strict Monotonicity Implies Injectivity:

• Every strictly monotone function is injective.

D Even and Odd Functions:

• Let $f : \mathbb{R} \rightarrow \mathbb{R}$.

• f is *even* if

$$f(-x) = f(x).$$

• f is *odd* if

$$f(-x) = -f(x).$$

D Convexity and Concavity (Geometric):

• The graph of a function is called

• *convex* if it bends to the left when moving from left to right,

• *concave* if it bends to the right when moving from left to right.

4.2 Limits of Functions

D Limit of a Function:

• Let $f : D(f) \rightarrow \mathbb{R}$ and $x_0 \in \mathbb{R}$. A number $L \in \mathbb{R}$ is called the limit of $f(x)$ at x_0 if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } x \in D(f), |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

The limit is uniquely determined.

D One-Sided Limits:

• One-Sided Limits

• Right-hand limit:

$$\lim_{x \rightarrow x_0^+} f(x) = L$$

• Left-hand limit:

$$\lim_{x \rightarrow x_0^-} f(x) = L$$

D Limits at Infinity:

• Limits at Infinity

$$\lim_{x \rightarrow \infty} f(x) = L$$

$$\lim_{x \rightarrow -\infty} f(x) = L$$

D Infinite Limits:

• Infinite Limits

$$\lim_{x \rightarrow c} f(x) = \infty$$

$$\lim_{x \rightarrow c} f(x) = -\infty$$

T Limit Laws:

• Let

$$\lim_{x \rightarrow c} f(x) = K, \quad \lim_{x \rightarrow c} g(x) = L, \quad K, L \neq \pm\infty.$$

Then:

$$\lim_{x \rightarrow c} (f + g) = K + L$$

$$\lim_{x \rightarrow c} (f - g) = K - L$$

$$\lim_{x \rightarrow c} (fg) = KL$$

$$\lim_{x \rightarrow c} (Af) = AK$$

$$\text{• If } L \neq 0: \lim_{x \rightarrow c} \frac{f}{g} = \frac{K}{L}$$

4.3 Continuity

D Continuity:

- Let $f : D \rightarrow \mathbb{R}$ and $x_0 \in D$. The function f is called *continuous at x_0* if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon.$$

The function is called *continuous* if it is continuous at every point in its domain.

D One-Sided Continuity:

- Let $f : D(f) \rightarrow \mathbb{R}$ and $x_0 \in D(f)$.
 - f is *right-continuous* at x_0 if

$$\lim_{x \rightarrow x_0^+} f(x) = f(x_0).$$

- f is *left-continuous* at x_0 if

$$\lim_{x \rightarrow x_0^-} f(x) = f(x_0).$$

T Sequential Characterization of Continuity:

- Let $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and $x_0 \in D$. Then f is continuous at x_0 if and only if for every sequence (x_n) with $x_n \rightarrow x_0$ we have

$$f(x_n) \rightarrow f(x_0).$$

T Algebra of Continuous Functions:

- Let $f, g : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Then:

- $f + g$ is continuous
- $f - g$ is continuous
- cf is continuous for any constant $c \in \mathbb{R}$
- $f \cdot g$ is continuous
- $\frac{f}{g}$ is continuous if $g \neq 0$

T Continuity of Composition:

- Let $f : I \rightarrow \text{image}(f)$ and $g : \text{image}(f) \rightarrow \mathbb{R}$ be continuous. Then the composition

$$g \circ f$$

is continuous and

$$\lim_{x \rightarrow x_0} g(f(x)) = g\left(\lim_{x \rightarrow x_0} f(x)\right).$$

T Continuity of the Inverse:

- Let I be an interval and let $f : I \rightarrow J$ be continuous and bijective. Then J is also an interval and the inverse function

$$f^{-1} : J \rightarrow I$$

is continuous.

T Intermediate Value Theorem:

- Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and let c be a value between $f(a)$ and $f(b)$. Then there exists $x \in [a, b]$ such that

$$f(x) = c.$$

C Zero of a Continuous Function (Bolzano theorem):

- Let $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be continuous and suppose

$$f(a) < 0, \quad f(b) > 0.$$

Then there exists $x \in (a, b)$ such that

$$f(x) = 0.$$

D Compact Interval:

- A bounded and closed interval $[a, b]$ is called *compact*.

L Bolzano–Weierstrass for Intervals:

- Let (x_n) be a sequence in a compact interval $[a, b]$. Then there exists a subsequence (x_{n_k}) such that

$$x_{n_k} \rightarrow x_0 \quad \text{with } x_0 \in [a, b].$$

D Local Extrema:

- Let $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$.

- x_0 is a *local maximum* if

$$f(x_0) \geq f(x) \quad \forall x \in D.$$

- x_0 is a *local minimum* if

$$f(x_0) \leq f(x) \quad \forall x \in D.$$

- Local maxima and minima are called *local extrema*.

T Extreme Value Theorem:

- Let $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be continuous and let I be compact. Then f is bounded and attains its maximum and minimum on I , i.e., there exist $x_{\min}, x_{\max} \in I$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \forall x \in I.$$

5 Derivatives

5.1 Differentiation

D Derivative:

• Let $f : D \rightarrow \mathbb{R}$ and assume D has no isolated points. The *difference quotient* is

$$\frac{f(a+h) - f(a)}{h}.$$

If the limit exists, the *derivative* of f at a is defined by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a}.$$

Interpretation:

- instantaneous rate of change
- slope of the tangent
- sign indicates increase/decrease

D Derivative Function:

• Given $f : D \rightarrow \mathbb{R}$, the *derivative function* is

$$x \mapsto f'(x).$$

D Differentiability:

• A function f is *differentiable at x* if $f'(x)$ exists. If this holds for all x in the domain, f is called differentiable.

L Differentiability Implies Continuity:

• If f is differentiable at x_0 , then f is continuous at x_0 .

D One-Sided Derivatives:

•

• Right derivative:

$$\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$$

• Left derivative:

$$\lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}$$

D Higher Derivatives:

• Define iteratively:

$$f^{(0)} = f, \quad f^{(1)} = f', \quad f^{(2)} = f'', \quad f^{(n+1)} = (f^{(n)})'.$$

If $f^{(n)}$ exists, f is n -times differentiable. If all derivatives up to order n are continuous, then $f \in C^n(D)$. Define

$$C^\infty(D) := \bigcap_{n=0}^{\infty} C^n(D),$$

called *smooth functions*.

T Basic Derivatives:

•

• Power:

$$(ax^p)' = apx^{p-1}$$

• Exponential:

$$(a^x)' = \ln(a) a^x$$

• Trigonometric:

$$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x, \quad (\tan x)' = \frac{1}{\cos^2 x}$$

• Hyperbolic:

$$(\sinh x)' = \cosh x, \quad (\cosh x)' = \sinh x$$

• Inverse:

$$(\ln x)' = \frac{1}{x}, \quad (\arctan x)' = \frac{1}{1+x^2}$$

T Sum and Product Rule:

• For differentiable f, g :

$$(f+g)' = f' + g'$$

$$(fg)' = f'g + fg'$$

T Chain Rule:

• If f is differentiable at x_0 and g at $f(x_0)$, then

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

T Quotient Rule:

• If $g(x_0) \neq 0$, then

$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}.$$

T Derivative of the Inverse:

• Let f be bijective and differentiable with $f'(x_0) \neq 0$. Then

$$(f^{-1})'(f(x_0)) = \frac{1}{f'(x_0)}.$$

T Necessary Condition for Extremum:

• If x_0 is a local extremum and f is differentiable at x_0 , then

$$f'(x_0) = 0.$$

T Characterization of Local Extrema:

• If x_0 is a local extremum, then at least one holds:

- x_0 is a boundary point

• f is not differentiable at x_0

• $f'(x_0) = 0$

T Rolle's Theorem:

• Let f be continuous on $[a, b]$ and differentiable on (a, b) with $f(a) = f(b)$. Then there exists $\xi \in (a, b)$ such that

$$f'(\xi) = 0.$$

T Mean Value Theorem:

• Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $\xi \in (a, b)$ such that

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

T l'Hospital's Rule (0/0):

• If

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$$

and

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$$

T l'Hospital's Rule (∞/∞):

• If

$$\lim_{x \rightarrow a} |f(x)| = \lim_{x \rightarrow a} |g(x)| = \infty$$

and

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$$

T l'Hospital at Infinity:

• If

$$\lim_{x \rightarrow \infty} f(x), g(x) \in \{0, \infty\}$$

and

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L.$$

T Monotonicity via Derivative:

•

$$f'(x) \geq 0 \iff f \text{ is increasing}$$

C Constant Functions:

• f is constant iff

$$f'(x) = 0 \quad \forall x.$$

T Convexity via First Derivative:

- f is convex iff f' is increasing.

C Convexity via Second Derivative:

- If f is twice differentiable:

$$f''(x) \geq 0 \iff f \text{ is convex.}$$

$$f''(x) \leq 0 \iff f \text{ is concave}$$

T Second Derivative Test:

- Let f be twice differentiable near x_0 and suppose $f'(x_0) = 0$.

Then: $f''(x_0) > 0 \Rightarrow x_0$ is a local minimum,

and $f''(x_0) < 0 \Rightarrow x_0$ is a local maximum.

5.2 Taylor Approximation and Power Series**Local Approximation by Polynomials:**

- In a sufficiently small neighborhood of a point x_0 , the graph of a differentiable function f behaves almost like its tangent at x_0 . Hence, the function can locally be approximated by its tangent.

D Linear Approximation / Tangent Approximation:

- If f is differentiable at x_0 , then

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

The expression

$$P_1(x) = f(x_0) + f'(x_0)(x - x_0)$$

is called the *linearization*, *tangent approximation*, or *first Taylor polynomial* of f at x_0 .

T Taylor's Theorem:

- Let f be infinitely differentiable on an open interval I containing x_0 . Then for every $x \in I$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{1}{n!} f^{(n)}(x_0)(x - x_0)^n,$$

for some c between x_0 and x . Equivalently,

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x),$$

where

$$R_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c)(x - x_0)^{n+1}.$$

The polynomial

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

is called the n -th *Taylor polynomial*.

Taylor Approximation and Power Series:

- Taylor polynomials provide increasingly accurate approximations of

many functions. For many functions, taking infinitely many terms recovers the original function.

D Taylor Series:

- For a function f , the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k$$

is called the *Taylor series* (or Taylor expansion) of f around the point a .

T Important Taylor Series:

- The following series converge for all x :

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

T Further Important Power Series:

- The following series converge for $-1 < x < 1$.

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$$

$$(1+x)^p = \sum_{n=0}^{\infty} \binom{p}{n} x^n = 1 + px + \frac{p(p-1)}{2!} x^2 + \cdots$$

Operations on Taylor Series:

- New Taylor series can be constructed from known ones using:
 - substitution
 - multiplication
 - differentiation
 - integration

T Multiplication of Power Series:

- Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad g(x) = \sum_{n=0}^{\infty} b_n x^n,$$

with x inside both convergence intervals. Then

$$f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n.$$

T Termwise Differentiation:

- Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

inside its convergence interval. Then f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

T Termwise Integration:

- Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

inside its convergence interval I . Then for every interval $[a, b] \subseteq I$,

$$\int_a^b f(x) dx = \sum_{n=0}^{\infty} \int_a^b a_n x^n dx.$$

6 Differential equations

D Differential Equation:

- A *differential equation* is an equation involving an unknown function together with its derivatives.

$$u'(t) = r(a - u(t))(b - u(t))$$

where r, a, b are constants, another example

$$y^{(4)}(x) - 4y^{(3)}(x) + 5y''(x) - 4y'(x) + 4y(x) = 0.$$

The variables t, x are called *independent variables*, while u, y are called *dependent variables*.

D Linear Differential Equation:

- A differential equation of the form

$$a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \cdots + a_1(x)y'(x) + a_0(x)y(x) = s(x)$$

is called a *linear differential equation* with coefficients $a_i(x)$.

The *order* of the differential equation is the highest order derivative appearing in the equation. The function $s(x)$ is called the *forcing term*. If $s(x) = 0$, the equation is called *homogeneous*; otherwise it is called *inhomogeneous*.

D Differential Equation with Constant Coefficients:

- If all coefficient functions $a_i(x)$ are constant functions, the equation is called a *differential equation with constant coefficients*.

T Superposition Principle:

- If $y_1(x)$ and $y_2(x)$ are solutions of a linear homogeneous differential equation, then

$$y(x) = C_1 y_1(x) + C_2 y_2(x), \quad C_1, C_2 \in \mathbb{R},$$

is also a solution of the same differential equation.

T Fundamental Solutions:

- A linear homogeneous differential equation of order n possesses n solutions $y_1(x), \dots, y_n(x)$ such that the general solution is given by

$$y(x) = C_1 y_1(x) + \cdots + C_n y_n(x), \quad C_1, \dots, C_n \in \mathbb{R}.$$

These solutions are called *fundamental solutions*.

Characteristic Polynomial:

- Consider the differential equation

$$a_n u^{(n)}(x) + a_{n-1} u^{(n-1)}(x) + \cdots + a_1 u'(x) + a_0 u(x) = 0.$$

Using the ansatz $u(x) = e^{\lambda x}$ (Euler ansatz), we obtain the *characteristic equation*

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \cdots + a_1 \lambda + a_0 = 0.$$

We get the *characteristic polynomial*:

$$p(\lambda) = a_n \lambda^n + a_{n-1} \lambda^{n-1} + \cdots + a_1 \lambda + a_0$$

T Solution of Linear Homogeneous Differential Equations:

- Consider the differential equation

$$a_n u^{(n)}(x) + a_{n-1} u^{(n-1)}(x) + \cdots + a_1 u'(x) + a_0 u(x) = 0, \quad a_i \in \mathbb{R}.$$

If all roots λ_i of the characteristic polynomial are real and have multiplicities m_i , then $u(x) = \sum_{i=1}^r \left(\sum_{p=0}^{m_i-1} C_{ip} x^p e^{\lambda_i x} \right)$.

If there are additionally complex conjugate roots $a_j \pm ib_j$ with multiplicities m_j , then $u(x) = \sum_{i=1}^r \left(\sum_{p=0}^{m_i-1} C_{ip} x^p e^{\lambda_i x} \right) + \sum_{j=1}^s \left(\sum_{q=0}^{m_j-1} A_{jq} x^q e^{a_j x} \sin(b_j x) + B_{jq} x^q e^{a_j x} \cos(b_j x) \right)$.

Constants in the General Solution:

- The general solution of a differential equation of order n contains n constants. Therefore, in order to determine a unique solution, one must specify n additional conditions.

D Boundary Value Problem:

- Conditions imposed at different points, e.g. for $n = 2$:

$$u(t_0) = u_0, \quad u(t_1) = u_1, \quad t_0 \neq t_1,$$

define a *boundary value problem*.

D Initial Value Problem:

- Conditions imposed at a single point, e.g. for $n = 2$:

$$u(t_0) = u_0, \quad u'(t_0) = v_0,$$

define an *initial value problem*.

7 Integration

8 Integration

D Antiderivative / Indefinite Integral:

- Let $I \subseteq \mathbb{R}$ be an interval and let

$$f : I \rightarrow \mathbb{R}$$

be a function. A differentiable function

$$F : I \rightarrow \mathbb{R}$$

satisfying

$$F'(x) = f(x) \quad \forall x \in I$$

is called an *antiderivative* of f on I . Determining F from f is called *indefinite integration*.

Non-Uniqueness of Antiderivatives:

- An antiderivative is not unique. If $F(x)$ is an antiderivative of $f(x)$, then

$$F(x) + C$$

is also an antiderivative for every constant $C \in \mathbb{R}$. Hence a function possesses an entire family of antiderivatives. This family is called the *indefinite integral* and is written as

$$\int f(x) dx.$$

The function f is called the *integrand*.

T Linearity of the Indefinite Integral:

- For indefinite integrals:

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

and

$$\int s f(x) dx = s \int f(x) dx,$$

where s is a constant.

T Basic Integration Rules:

- Rules:

- $\int k dx = kx + C$
- $\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1$
- $\int e^x dx = e^x + C$
- $\int \frac{1}{x} dx = \ln(|x|) + C$
- $\int \sin(x) dx = -\cos(x) + C$
- $\int \cos(x) dx = \sin(x) + C$

- $\int \sinh(x) dx = \cosh(x) + C$

- $\int \cosh(x) dx = \sinh(x) + C$

Differentiation Rules vs. Integration Rules:

- Recall the two fundamental differentiation rules:

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

(chain rule) and

$$\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

(product rule). The goal is to derive and apply the corresponding rules in integration theory.

T Integration by Parts:

- Using the product rule:

$$\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x),$$

we obtain

$$\int \frac{d}{dx} (f(x)g(x)) dx = (fg)(x).$$

Therefore,

$$(fg)(x) = \int f'(x)g(x) dx + \int f(x)g'(x) dx.$$

Rearranging gives the formula for *integration by parts*:

$$\int f'(x)g(x) dx = f(x)g(x) - \int f(x)g'(x) dx$$

T Integration by Substitution:

- Using the chain rule:

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x),$$

we obtain

$$\int f'(g(x))g'(x) dx = f(g(x)) + C.$$

With the substitution

$$y = g(x),$$

this becomes

$$\int f'(g(x))g'(x) dx = \int \frac{df}{dy} dy.$$

Hence the substitution rule is

$$\int f'(g(x))g'(x) dx = f(g(x)) + C$$

9 Integration Part 2

Motivation:

- Given a velocity function $v(t)$ describing one-dimensional motion, natural questions are:

- What is the total distance traveled?

- What is the change in position?

D Partition of an Interval:

- Let $[a, b] \subseteq \mathbb{R}$ be a compact interval. A *partition* of $[a, b]$ is a finite set of points

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

The points x_i are called *subdivision points*. A partition

$$a = y_0 < y_1 < \cdots < y_m = b$$

is called a *refinement* of

$$a = x_0 < x_1 < \cdots < x_n = b$$

if

$$\{x_0, \dots, x_n\} \subseteq \{y_0, \dots, y_m\}.$$

D Step Function:

- A function

$$f : [a, b] \rightarrow \mathbb{R}$$

is called a *step function* if there exists a partition

$$a = x_0 < x_1 < \cdots < x_n = b$$

such that for every $k = 1, \dots, n$, the restriction

$$f|_{(x_{k-1}, x_k)}$$

is constant.

D Integral of a Step Function:

- Let

$$f : [a, b] \rightarrow \mathbb{R}$$

be a step function with respect to the partition

$$a = x_0 < x_1 < \cdots < x_n = b.$$

Then the integral of f over $[a, b]$ is defined by

$$\int_a^b f(x) dx := \sum_{k=1}^n c_k (x_k - x_{k-1}),$$

where c_k denotes the constant value of f on (x_{k-1}, x_k) .

D Upper and Lower Sums:

- Let

$$f : [a, b] \rightarrow \mathbb{R}.$$

The set of upper sums is defined by

$$U(f) := \left\{ \int_a^b s(x) dx \mid s \in TF, s \geq f \right\},$$

and the set of lower sums by

$$L(f) := \left\{ \int_a^b s(x) dx \mid s \in TF, s \leq f \right\},$$

where TF denotes the set of step functions.

D Riemann Integral:

- A bounded function

$$f : [a, b] \rightarrow \mathbb{R}$$

is called *Riemann integrable* if

$$\sup L(f) = \inf U(f).$$

In this case, the common value is called the *Riemann integral* and is denoted by

$$\int_a^b f(x) dx := \sup L(f) = \inf U(f).$$

The numbers a, b are called the lower and upper limits of integration, and f is called the integrand.

T Properties of the Riemann Integral:

- (i) (Linearity) If $f, g : [a, b] \rightarrow \mathbb{R}$ are integrable and $\alpha, \beta \in \mathbb{R}$, then

$$\int_a^b (\alpha f + \beta g)(x) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx.$$

- (ii) (Monotonicity) If

$$f(x) \leq g(x)$$

for all x , then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

- (iii) (Triangle Inequality)

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$$

- (iv) (Reversal of Integration Limits)

$$\int_a^b f(x) dx = - \int_b^a f(x) dx.$$

- (v) (Splitting the Interval) For $a \leq c \leq b$:

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

T Monotone Functions are Integrable:

- Every monotone function

$$f : [a, b] \rightarrow \mathbb{R}$$

is Riemann integrable.

T Continuous Functions are Integrable:

- Every continuous function

$$f : [a, b] \rightarrow \mathbb{R}$$

is Riemann integrable.

D Pointwise Convergence of Functions:

- Let

$$(f_n)_{n \in \mathbb{N}_0}$$

be a sequence of functions

$$f_n : D \rightarrow \mathbb{R},$$

and let

$$f : D \rightarrow \mathbb{R}.$$

The sequence (f_n) converges *pointwise* to f if for every $x \in D$, the sequence

$$(f_n(x))_{n \in \mathbb{N}_0}$$

converges to $f(x)$.

D Uniform Convergence:

- The sequence (f_n) converges *uniformly* to f if

$$\forall \varepsilon > 0 \exists N$$

such that for all

$$n \geq N \quad \text{and all } x \in D,$$

we have

$$|f_n(x) - f(x)| < \varepsilon.$$

T Uniform Limit of Continuous Functions:

- Let

$$(f_n)_{n \in \mathbb{N}_0}$$

be a sequence of continuous functions

$$f_n : D \rightarrow \mathbb{R}$$

converging uniformly to

$$f : D \rightarrow \mathbb{R}.$$

Then f is continuous.

T Uniform Limit and Integration:

- Let

$$(f_n)_{n \in \mathbb{N}_0}$$

be a sequence of integrable functions

$$f_n : [a, b] \rightarrow \mathbb{R}$$

converging uniformly to

$$f : [a, b] \rightarrow \mathbb{R}.$$

Then f is integrable and

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx.$$

T Mean Value Theorem for Integrals:

- If

$$f : [a, b] \rightarrow \mathbb{R}$$

is continuous, then there exists

$$c \in [a, b]$$

such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

The quantity

$$\frac{1}{b-a} \int_a^b f(x) dx$$

is called the *average value* of f on $[a, b]$.

T Fundamental Theorem of Calculus:

- Let

$$f : [a, b] \rightarrow \mathbb{R}$$

be continuous. Then for every constant $C \in \mathbb{R}$, the function

$$F(x) := \int_a^x f(y) dy + C$$

is an antiderivative of f . Moreover, every antiderivative of f is of this form.

C Integral Representation:

- Let

$$F : [a, b] \rightarrow \mathbb{R}$$

be continuously differentiable. Then

$$F(x) = F(a) + \int_a^x F'(t) dt.$$

C Newton–Leibniz Formula:

- Let

$$f : [a, b] \rightarrow \mathbb{R}$$

be continuous and let F be an antiderivative of f . Then

$$\int_a^b f(t) dt = F(b) - F(a).$$

T Integration by Parts:

- Let

$$f, g : [a, b] \rightarrow \mathbb{R}$$

be continuously differentiable. Then

$$\int_a^b f(x)g'(x) dx = f(x)g(x) \Big|_a^b - \int_a^b f'(x)g(x) dx.$$

T Substitution Rule (Version 1):

- Let

$$f : I \rightarrow J$$

be continuously differentiable and

$$g : J \rightarrow \mathbb{R}$$

continuous. Then for every

$$[a, b] \subseteq I,$$

$$\int_a^b g(f(x))f'(x) dx = \int_{f(a)}^{f(b)} g(y) dy.$$

T Substitution Rule (Version 2):

- Let

$$f : I \rightarrow J$$

be continuously differentiable and

$$g : J \rightarrow \mathbb{R}$$

continuous. Assume

$$f'(x) \neq 0 \quad \forall x \in [a, b]$$

and let

$$f^{-1} : [f(a), f(b)] \rightarrow \mathbb{R}$$

denote the inverse of $f|_{[a,b]}$. Then

$$\int_a^b g(f(x)) dx = \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y) dy.$$

Important Limits

$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
$\lim_{x \rightarrow \infty} e^x = \infty$	$\lim_{x \rightarrow -\infty} e^x = 0$
$\lim_{x \rightarrow \infty} e^{-x} = 0$	$\lim_{x \rightarrow -\infty} e^{-x} = \infty$
$\lim_{x \rightarrow \infty} \ln x = \infty$	$\lim_{x \rightarrow 0} \ln x = -\infty$
$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$	$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$	$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$
$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$	$\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$
$\lim_{x \rightarrow 0^+} x \ln x = 0$	$\lim_{x \rightarrow \infty} n^{1/n} = 1$
$\lim_{x \rightarrow \pm \infty} \left(1 + \frac{1}{x}\right)^x = e$	$\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^x = \frac{1}{e}$
$\lim_{x \rightarrow \pm \infty} \left(1 + \frac{k}{x}\right)^{mx} = e^{km}$	$\lim_{x \rightarrow \infty} x^a q^x = 0$

Rules of derivation

- **Sum:** $(f + g)' = f' + g'$
- **Product:** $(fg)' = f'g + fg'$
- **Chain:** $(g \circ f)' = g'(f)f'$
- **Quotient:** $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
- **Derivative of the Inverse:** $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}, \quad y_0 = f(x_0)$

Rules of integration

- **Linearity of Integration:** $\int (f + g) = \int f + \int g, \quad \int cf = c \int f$
- **Integration by Parts:** $\int f'g = fg - \int fg'$
- **Substitution:** $\int g(f(x))f'(x) \, dx = \int g(y) \, dy$

Logarithms

- **Change of base:** $\log_a x = \frac{\ln x}{\ln a}$
- **Power rule:** $\log_a (x^y) = y \log_a x$
- **Product rule:** $\log_a (xy) = \log_a x + \log_a y$
- **Quotient rule:** $\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$
- **Logarithm 1:** $\log_a (1) = 0 \, \forall a \in \mathbb{N}$ (negative undefined)

Trigonometry

- **Tangent and cotangent:** $\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}$
- **Pythagorean identity:** $\sin^2 x + \cos^2 x = 1$
- **Double-angle sine:** $\sin(2x) = 2 \sin x \cos x$
- **Double-angle cosine:** $\cos(2x) = \cos^2 x - \sin^2 x$
- **Angle addition sine:** $\sin(x + y) = \sin x \cos y + \cos x \sin y$
- **Angle addition cosine:** $\cos(x + y) = \cos x \cos y - \sin x \sin y$
- **Even/Odd symmetry:** $\cos(x) = \cos(-x)$ and $\sin(-x) = -\sin(x)$
- **Reflection at $\pi/2$:** $\cos(\pi - x) = -\cos(x)$ and $\sin(\pi - x) = \sin(x)$

Hyperbolic

- **Hyperbolic sine:** $\sinh(x) := \frac{e^x - e^{-x}}{2} : \mathbb{R} \rightarrow \mathbb{R}$
- **Hyperbolic cosine:** $\cosh(x) := \frac{e^x + e^{-x}}{2} : \mathbb{R} \rightarrow [1, \infty)$
- **Hyperbolic tangent:** $\tanh(x) := \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} : \mathbb{R} \rightarrow (-1, 1)$
- **Inverse hyperbolic sine:** $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$
- **Inverse hyperbolic cosine:** $\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$
- **Inverse hyperbolic tangent:** $\operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$

Complement trick

$$\sqrt{ax + b} - \sqrt{cx + d} = \frac{ax + b - (cx + d)}{\sqrt{ax + b} + \sqrt{cx + d}}$$

Derivatives and Antiderivatives

Antiderivative	Function	Derivative
$\frac{x^{n+1}}{n+1}$	x^n	nx^{n-1}
$\ln x $	$\frac{1}{x} = x^{-1}$	$-x^{-2} = -\frac{1}{x^2}$
$\frac{2}{3}x^{3/2}$	$\sqrt{x} = x^{1/2}$	$\frac{1}{2\sqrt{x}}$
$\frac{n}{n+1}x^{1/n+1}$	$\sqrt[n]{x} = x^{1/n}$	$\frac{1}{n}x^{1/n-1}$
e^x	e^x	e^x
$\exp(x)$	$\exp(x)$	$\exp(x)$
$\frac{1}{a(n+1)}(ax + b)^{n+1}$	$(ax + b)^n$	$n(ax + b)^{n-1}a$
$x(\ln x - 1)$	$\ln(x)$	$\frac{1}{x}$
$\frac{1}{\ln(a)}a^x$	a^x	$a^x \ln(a)$
$\frac{x}{\ln(a)}(\ln x - 1)$	$\log_a x $	$\frac{1}{x \ln(a)}$
$-\cos(x)$	$\sin(x)$	$\cos(x)$
$\sin(x)$	$\cos(x)$	$-\sin(x)$
$-\ln \cos(x) $	$\tan(x)$	$\frac{1}{\cos^2(x)}$
$\ln \sin(x) $	$\cot(x)$	$-\frac{1}{\sin^2(x)}$
$x \arcsin(x) + \sqrt{1 - x^2}$	$\arcsin(x)$	$\frac{1}{\sqrt{1 - x^2}}$
$x \arccos(x) - \sqrt{1 - x^2}$	$\arccos(x)$	$-\frac{1}{\sqrt{1 - x^2}}$
$x \arctan(x) - \frac{1}{2} \ln(x^2 + 1)$	$\arctan(x)$	$\frac{1}{x^2 + 1}$
$\ln \sin(x) $	$\cot(x)$	$-\frac{1}{\sin^2(x)}$
$\cosh(x)$	$\sinh(x)$	$\cosh(x)$
$\sinh(x)$	$\cosh(x)$	$\sinh(x)$
$\ln \cosh(x) $	$\tanh(x)$	$\frac{1}{\cosh^2(x)}$
	$\operatorname{arsinh}(x)$	$\frac{1}{\sqrt{1 + x^2}}$
	$\operatorname{arcosh}(x)$	$\frac{1}{\sqrt{x^2 - 1}}$
	$\operatorname{artanh}(x)$	$\frac{1}{1 - x^2}$

Special Trigonometric Values

°	rad	$\sin(\xi)$	$\cos(\xi)$	$\tan(\xi)$
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	∅
120°	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
135°	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
150°	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
180°	π	0	-1	0

Further Derivatives

$F(x)$	$f(x) = F'(x)$
$\frac{1}{a} \ln ax + b $	$\frac{1}{ax + b}$
$\frac{ax}{c} - \frac{ad - bc}{c^2} \ln cx + d $	$\frac{a(cx + d) - c(ax + b)}{(cx + d)^2}$
$\frac{x}{2} \sqrt{a^2 + x^2} + \frac{a^2}{2} \ln x + \sqrt{a^2 + x^2} $	$\sqrt{a^2 + x^2}$
$\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin\left(\frac{x}{a}\right)$	$\sqrt{a^2 - x^2}$
$\frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln x + \sqrt{x^2 - a^2} $	$\sqrt{x^2 - a^2}$
$\ln x + \sqrt{x^2 \pm a^2} $	$\frac{1}{\sqrt{x^2 \pm a^2}}$
$\arcsin\left(\frac{x}{a}\right)$	$\frac{1}{\sqrt{a^2 - x^2}}$
$\frac{1}{a} \arctan\left(\frac{x}{a}\right)$	$\frac{1}{a^2 + x^2}$
$-\frac{1}{a} \frac{\cos(ax + b) \sin(ax + b) \frac{1}{a} \sin(ax + b) \cos(ax + b)}{x^x x^x (1 + \ln x) (x^x)^x (x^x)^x (x + 2x \ln x) x^x x^x (x^{x-1} + x^x (1 + \ln x) \ln x)}$	height